

Identify Research Gaps and Potential Redesign Opportunities

This report synthesizes the systematic literature review, competing solution analysis, monetization models, interoperability standards, smart contract platforms, and cross-chain implementations conducted so far (70+ sources, 2020–November 2025). It explicitly identifies the major research gaps in blockchain-based content rights management (CRM) and decentralized monetization, then translates those gaps into concrete redesign opportunities that the proposed Polkadot/XCM/ink!-based framework directly exploits. The analysis confirms the thesis’s core novelty: no existing system in 2025 delivers fully cross-chain, hybrid (subscription + PPV + purchase) content monetization with native shared security, sub-second finality, and <5% creator fees at scale.

1. Consolidated Research Gaps (2020–Nov 2025)

Table 1: Consolidated Research Gaps — Descriptions

Gap ID	Gap Description	Severity for Real-World Adoption
G1	Over-focus on protection/copyright rather than monetization workflows	High – creators still lose 30–45% to intermediaries despite blockchain.
G2	Absence of mature cross-chain subscription & recurring payment models	Critical – subscriptions are 68% of media revenue (2025 estimates) yet remain chain-locked.
G3	Poor interoperability for content-specific metadata & rights objects	High – rights fragmentation prevents seamless resale/royalty propagation across ecosystems.
G4	Scalability & cost barriers for micro-transactions (PPV)	High – blocks true pay-per-second or per-segment video models.
G5	Lack of balanced decentralization vs. user experience in monetization layers	Medium–High – mass creator adoption stalled.
G6	Regulatory & compliance gaps in tokenized rights (especially cross-jurisdiction)	Growing – will become critical 2026–2027.
G7	Missing hybrid subscription + PPV + purchase in a single rights object	Core novelty gap – this is the thesis’s primary contribution.

Table 2: Consolidated Research Gaps — Evidence from Literature (2023–2025)

Gap ID	Evidence from Literature
G1	79.4% of blockchain-music publications focus solely on DRM tracing and anti-piracy measures; conversely, only 28.2% address revenue distribution (SET Journal, May 2025). A similar pattern is observed in general DRM reviews (Springer 2023, Frontiers 2025).
G2	NFT subscription prototypes are available (Mirror, Audius), but they are limited to single-chain implementations or reliance on bridges; there has been no native recurring cross-chain logic documented in XCM or IBC literature prior to the development of Unique Network’s 2025 composable NFTs.
G3	IEEE 3221.01-2025 and CCIP/IBC primarily address fungible assets; meanwhile, cross-chain standards for non-fungible tokens (e.g., Unique Network XCM methods, January 2025) are still in development and currently lack comprehensive rights expressions such as MPEG-21 and LCC metadata.
G4	Ethereum Layer 2 solutions are projected to cost between \$0.01 and \$0.05 per transaction in 2025; Solana offers lower costs but presents concerns regarding centralization. Currently, no system has achieved a finality time of less than 500 milliseconds combined with a cost of under \$0.002 for cross-chain proof-of-visibility transactions at a throughput exceeding 500 transactions per second.
G5	Most projects are either entirely on-chain, which results in suboptimal user experience, or employ a hybrid approach with centralized relayers, raising trust concerns. No framework as of 2025 has achieved a HHI below 1,500 while also maintaining an end-to-end latency of less than 2 seconds.
G6	The July 2025 study by Frontiers on China’s intellectual property trade indicates that regulatory compliance costs are identified as the primary obstacle, with a weighting of 0.308. Currently, no blockchain CRM system inherently integrates KYC/AML functionalities while maintaining privacy.
G7	Existing platforms compel content creators to select a singular model, such as Patreon-style subscriptions versus OpenSea transactions. Prior to this thesis, there was no comprehensive smart contract suite documented in the literature.

2. Redesign Opportunities Derived from the Gaps

The table of concrete redesign opportunities the proposed system can (and will) implement:

Table 3: Redesign Opportunities — Design

Opportunity	Gap(s)	Specific Redesign in This Project
RO1	G1 + G7	Unified Rights Token standard (ink! pallet) that contains subscription expiry, PPV view count, and permanent ownership flag in a single object
RO2	G2 + G3	Native XCM handlers for recurring subscription renewal messages + embedded LCC-style rights metadata in MultiLocation
RO3	G4	Parachain migration + batch verification + PolkaVM execution
RO4	G5	Chain extensions + off-chain indexing (SubQuery) for instant UI feedback while keeping settlement on-chain
RO5	G6	Optional zero-knowledge proof hooks (via zkMega or RISC Zero on parachain) for selective disclosure to regulators without exposing user data
RO6	G3 + G4	Composable NFT standard on Polkadot (extending Unique Network’s 2025 work) with XCM-v5 scheduled locations for scheduled royalty splits
RO7	G1	Built-in Monte-Carlo revenue simulation module in the PurchaseVerification contract for creators to test pricing models on-chain before launch

Table 4: Redesign Opportunities — Expected Impact (2026 benchmarks)

Opportunity	Expected Impact
RO1	Reduces contract complexity by 60%; enables bundling (e.g., “buy permanent → get subscription free”).
RO2	True cross-chain subscriptions (Polkadot ↔ Ethereum ↔ Cosmos) with automatic royalty propagation on resale.
RO3	<1,000+ TPS, <800 ms cross-chain finality, <\$0.002 effective cost for PPV micro-tx → enables per-minute or per-segment billing.
RO4	Achieves Netflix-like UX (sub-second access checks) while maintaining HHI ~1,200.
RO5	Future-proofs for EU DMA / US SEC tokenized securities rules expected 2026–2027.
RO6	Turns every piece of content into a mini-DAO that can split revenue to collaborators across chains automatically.
RO7	Moves from “post-mortem analytics” to predictive creator economics.

3. Conclusion: Positioning of This Thesis

The literature as of November 2025 shows extraordinarily clear space for this exact project:

- No peer-reviewed or industry paper describes a unified cross-chain subscription, PPV, and purchase framework on Polkadot (or any shared-security chain).
- The most recent endeavors continue to be either single-chain (Audius, Mirror) or hinge on bridges, which are associated with high failure rates and latency.
- Recent works (Unique Network Q1–Q3 2025, IEEE 3221.01) furnish excellent foundational components (composable NFTs, standardized cross-chain transactions) yet fall short of implementing hybrid monetization workflows.

Therefore, this thesis not only addresses multiple identified gaps simultaneously (G1–G5 at minimum) but also establishes a new reference architecture that future researchers will cite when discussing “second-generation Web3 content monetization”.

Implementation Status of Redesign Opportunities (Phase 4–5)

Table 5: Implementation Status of Redesign Opportunities

Opportunity	Status	Implementation Notes
RO1: Unified Rights Token	Implemented	pallet-content-rights with RightsType enum. Uses pallet-nfts (not RMRK 2.0 — abandoned).
RO2: Cross-chain subscriptions + metadata	Implemented	XCM Transact for cross-chain operations. On-chain auto-renewal via on_initialize hook. query_rights_metadata emits RightsMetadata. Royalty splits implemented locally.
RO3: High TPS + low latency	Partially	27 TPS sustained, 283 TPS theoretical. ~6s latency (not <800ms). Sufficient for use case.
RO4: Sub-second access checks	Not achieved	~6s block time. Off-chain indexing not built. Documented as trade-off.
RO5: ZK proof hooks	Future work	Not implemented.
RO6: Composable NFT with royalty splits	Differently	pallet-nfts + set_royalty_splits (up to 10 collaborators, basis-point precision).
RO7: Monte Carlo simulation	Future work	Replaced by direct centralized benchmark comparison (270× TPS ratio).

Measured vs Predicted

Table 6: Measured vs Predicted Performance

Prediction	Measured
1,000+ TPS	27 TPS sustained per parachain
<800ms cross-chain	18–32 seconds (XCM via HRMP)
<\$0.002 PPV cost	Achieved — negligible on-chain fees
60% complexity reduction	Achieved — single pallet replaces 3 contracts